Amrita School of Engineering, Bengaluru
Department of ECE
B. Tech. AY 2025-2026 (VII Semester)
Synopsis – 23ECE498 & 23EAC498/Project Phase-1

Batch Number/Batch ID: 43

Name of the Students and Signature (One below the other):

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Name of Guide and Signature:

Dr. Sushmitha Vekkot

Name of the Domain or TAG:

AI

Title of the Project Phase-1:

ZeroShield AI: An Explainable Framework for Zero-Day Threat Detection

Project Phase-1 Objective (s) along with the sub-objectives:

Objectives

- To develop a behavior-based anomaly detection system capable of identifying previously unseen (zero-day) cyber threats from network and host logs, along with an Explainable AI (XAI)-powered Security Operations Center (SOC) assistant that uses Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) for intelligent threat analysis.

Sub-objectives

- To integrate the MITRE ATT&CK framework and threat intelligence sources for contextual threat reasoning and attack technique mapping.
- To generate transparent and auditable explanations for each detected threat, including supporting evidence and confidence scores.
- To evaluate the proposed framework using public intrusion detection datasets based on metrics such as Precision, Recall, F1-score, Accuracy, and False Positive Rate.
- To evaluate detection performance using public intrusion detection datasets based on Precision, Recall, F1-score, Accuracy, and False Positive Rate, including generalization testing on attack categories held out of training to assess zero-day-style detection capability.
- To evaluate false-positive reduction by comparing alert outcomes with and without the self-verifying agentic stage on the same alert set.
- To assess the quality and faithfulness of generated explanations through human evaluation, checking that cited evidence and mapped ATT&CK techniques genuinely support each verdict.

References:

1. Srinivas, Siddhant, et al. "AI-augmented SOC: A survey of LLMs and agents for security automation." *Journal of Cybersecurity and Privacy* 5.4 (2025)
2. Chhetri, M. B., Tariq, S., Singh, R., Jalalvand, F., Paris, C., & Nepal, S. (2024). "Towards Human-AI Teaming to Mitigate Alert Fatigue in Security Operations Centres."
3. Chhabra, Anshuman, et al. "Agentic AI security: Threats, defenses, evaluation, and open challenges." *IEEE Access* (2026).
4. Roy, Joyjit, and Samaresh Kumar Singh. "AgentSOC: A multi-layer agentic AI framework for security operations automation." *2026 IEEE 5th International Conference on AI in Cybersecurity (ICAIC)*. IEEE, 2026.

Time Plan of Project Phase-1:

Month	Work to be implemented
July 2026	Literature survey on Zero-Day attack detection, Security Operations Centers (SOC), Explainable AI, RAG systems, LLM agents, and analysis of existing IDS approaches. Problem formulation and requirement analysis.
August 2026	Collection and preprocessing of cybersecurity datasets. Development of the anomaly detection pipeline and feature engineering.
September 2026	Development of the Retrieval-Augmented Generation (RAG) framework. Initial integration with the anomaly detection model.
October 2026	Design and implementation of the self-verifying multi-agent framework for alert validation. Development of explainable reasoning generation and confidence estimation.
November 2026	Integration of all modules into the ZeroShield AI prototype. Experimental evaluation, performance analysis, documentation, and preparation for Project Phase-1 review.